

Topic 4: Networks

Most computer applications in use today would not be possible without networks. Students should understand the key principles behind the organisation of computer networks.

Subject content	Students should:
4.1 Networks	4.1.1 understand why computers are connected in a network
	4.1.2 understand different types of networks (LAN, WAN)
	4.1.3 understand how the internet is structured (IP addressing, routers)
	4.1.4 understand how the characteristics of wired and wireless connectivity impact on performance (speed, range, latency, bandwidth)
	4.1.5 understand that network speeds are measured in bits per second (kilobit, megabit, gigabit) and be able to construct expressions involving file size, transmission rate and time
	4.1.6 understand the role of and need for network protocols (Ethernet, Wi-Fi, TCP/IP, HTTP, HTTPS, FTP) and email protocols (POP3, SMTP, IMAP)
	4.1.7 understand how the 4-layer (application, transport, internet, link) TCP/IP model handles data transmission over a network
	4.1.8 understand characteristics of network topologies (bus, star, mesh)